

GreenMorph: Sustainable Neuromorphic Computing through Energy-Harvesting and Energy-Driven Online STDP Learning

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Abstract

In recent years, the rapidly increasing computational demands of artificial intelligence (AI) have underscored the need for more sustainable and energy-efficient computing solutions. This paper introduces GreenMorph — a *carbon-aware Spiking Neural Network (SNN) framework* that relies on energy harvesting and can be driven by the energy level. GreenMorph dynamically adjusts timesteps and employs similarity-based neuron pruning to optimize energy consumption to achieve carbon neutrality during operation. Experimental evaluations on benchmark datasets show that timestep adaptation reduces training energy by 15.15% without accuracy degradation, while combining timestep adaptation with neuron pruning achieves an 88.10% reduction in training energy at the cost of only 3.01% accuracy loss. During inference, similarity-based neuron pruning reduces computation energy by 27.18% with just 0.35% accuracy loss, and by up to 88.32% with 8.81% accuracy loss.

1 Introduction

Over the past decade, the use of Artificial Intelligence (AI) in society has grown rapidly, with applications ranging from large language models to autonomous driving [1]. However, AI models incur significant energy consumption and associated carbon emissions. The carbon footprint of AI can be attributed to four main phases: manufacturing, transportation, operation, and recycling [2]. Among these, the operational phase contributes up to 70% of total emissions and thus offers the greatest potential for reduction [3]. Given its substantial impact and the challenges in reducing emissions in other phases, minimizing carbon emissions during the operational phase represents the most practical and effective approach toward sustainable AI.

Optimizing the operational phase requires both minimizing energy consumption and increasing the use of renewable energy. Conventional energy-efficient strategies primarily include algorithmic techniques such as Spiking Neural Networks (SNNs) [4], quantization [5], pruning [6], approximation [7], and compression [8]. In parallel, hardware-level approaches like power gat-

ing [9], clock gating [10, 11], and dynamic voltage and frequency scaling (DVFS) [12] have proven effective in reducing power consumption. Additionally, integrating renewable energy sources such as solar or wind can further decrease carbon emissions by up to 94.92% [13].

Despite the ability to reduce carbon emissions during the operational phase, significant challenges remain for edge and IoT devices, where energy supply is inherently limited. Most of these challenges arise due to power fluctuations, as power harvesting depends heavily on external factors such as weather and environmental conditions [14]. Figure 1 illustrates an example of solar energy fluctuation [14]. A significant difference in the weather will affect the ability to compute the system. Furthermore, as energy is stored in batteries, their capacity and voltage degrade over time, further reducing the reliability of the power supply [15] (see Figure 1(b)).

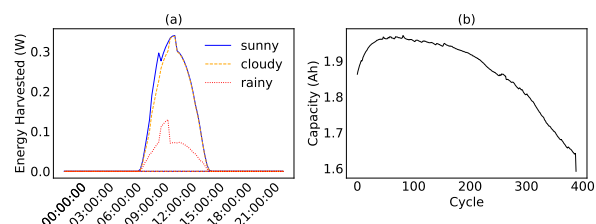


Figure 1: Example of solar energy fluctuation: (a) estimated power harvested using a 1-Watt solar panel under various weather conditions; (b) battery capacity degradation over charging and discharging cycles.

Therefore, there are main challenges in a carbon-neutral AI system:

- First, as noted above, energy harvesting (e.g., solar, wind) inherently fluctuates depending on external factors.
- Second, to address energy supply fluctuations, integrating low-power techniques may lead to potential errors. Although neural networks are somewhat tolerant of errors, training with potential errors needs further investigation.

To overcome these challenges, this work proposes an energy-efficient AI framework aimed at achieving

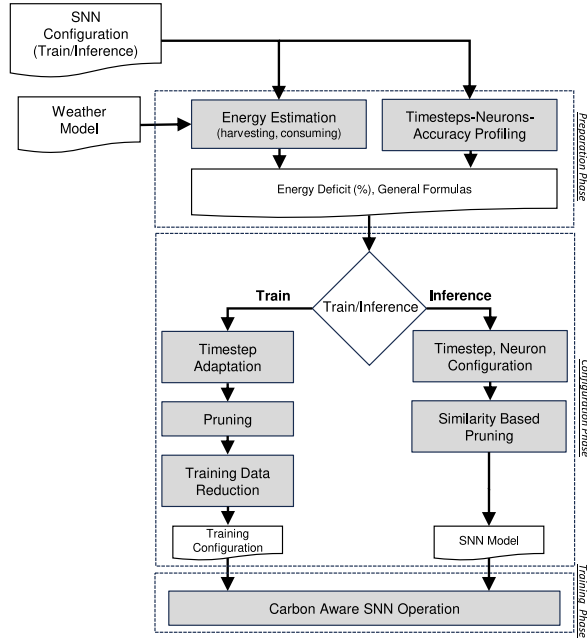


Figure 2: The flow of GreenMorph: SNN configuration, Weather, and Hardware configuration are taken into account to generate the training configuration or SNN model.

carbon-neutral (or net-zero) AI operations, including training and inference. The contributions of this work are summarized as follows:

- An optimization method that profiles the timestep-accuracy and neuron-accuracy trade-offs and generates a model configuration to achieve carbon-neutrality in the training.
- Similarity-based neuron pruning combined with timestep reduction to achieve carbon-neutrality in the inference.

The remainder of this paper is organized as follows. Section 2 describes the proposed framework. Section 3 details the evaluation methodology and results. Finally, Section 4 concludes the paper.

2 Proposed Framework

In this section, we will discuss the proposed GreenMorph framework and how it adapts the energy consumption to the energy supply. Figure 2 illustrates the proposed framework, which takes the following inputs: (1) SNN configuration, (2) weather model/condition, and hardware configuration (e.g., CPU/GPU, battery status, etc.). The framework's output is the adapted

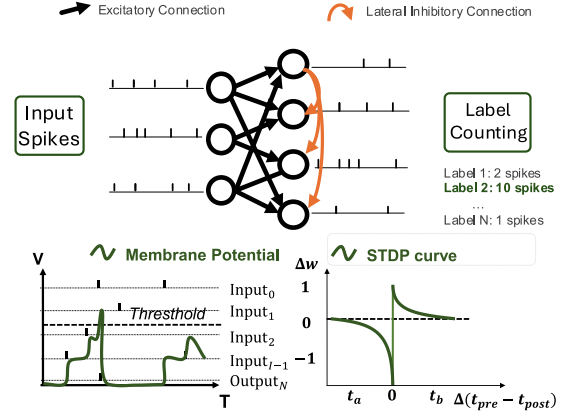


Figure 3: The STDP-based Spiking Network Model with lateral inhibitory connection.

training configuration or adapted pre-trained SNN model. Based on the output, SNN model training or inference is performed. The framework has three major phases: (1) preparation; (2) configuration, and (3) operation. Each phase is detailed in the following sections.

Figure 3 illustrates the neuromorphic model for STDP learning adopted from [16] with lateral inhibitory connections instead of a dedicated layer. Synaptic weights are trained with the STDP learning rule based on the relative time distance between the issuing timesteps of presynaptic and postsynaptic.

2.1 Preparation Phase

In the preparation phase, the framework prepares two important pieces of information: energy estimation and model timestep-accuracy, neuron-accuracy trade-offs. The amount of energy harvested is estimated based on open-sourced quartz-solar-forecast [14] developed by Open-Climate-Fix. The initial energy consumption is measured based on the given SNN configuration, including the number of training images. Based on the energy harvested $E_{harvested}$ and the initial energy consumption E_{init} , the energy deficit (ED) in percent, which indicates how much the system needs to reduce the energy consumption, is calculated:

$$ED = \left(\frac{E_{init} - E_{harvested}}{E_{init}} \right) \times 100\% \quad (1)$$

The preparation phase also derives general formulas based on the Timesteps-Neurons-Accuracy trade-offs. Since the effects of the number of timesteps and neurons differ between training and inference, different general formulas are used for training and inference. For

training, the number of timesteps is calculated as:

$$T_{timesteps} = a \times E_{train} + b \quad (2)$$

Where $T_{timesteps}$ and E_{train} are the number of timesteps and corresponding training energy, and a, b are coefficients. Using curve fitting based on the Levenberg-Marquardt method, Eq. 2 is fitted to the collected data points $(T_{timesteps}^*, E_{train}^*)$. Similarly, energy consumption in relation to the number of neurons is calculated as:

$$N_{neurons} = a \times E_{train}^d + b \times E_{train} + c \quad (3)$$

As the relation between energy consumption and the number of neurons is often non-linear, a polynomial formula is chosen. Eq. 3 is then fitted to the collected data points $(N_{neurons}^*, E_{train}^*)$.

For inference, $T_{timestep}$ and $N_{neurons}$ are simultaneously mapped from the inference energy using the following formula:

$$E_{inference} = aN_{neurons}^d + bT_{timestep} + c \quad (4)$$

Where $E_{inference}$ is the predicted inference energy consumption based on $N_{neurons}$ and $T_{timestep}$, and a, b, c, d are coefficients. The curve fitting process to data points $(N_{neurons}^*, T_{timestep}^*, E_{inference}^*)$ is as follows:

$$\min_{a,b,c,d} \sum_{i=1}^M [E_{inference,i} - E(N_{neurons,i}, T_{timestep,i})]^2 \quad (5)$$

Where M is the number of data points. After the coefficients for the approximation formula are found, we perform the optimization process with Sequential-Least-Squares-Programming (SLSQP) to map a given energy target (E_{target}) to a pair of $(N_{neurons}, T_{timestep})$.

$$\min_{N_{neurons}, T_{timestep}} [E_{inference} - E(N_{neurons}, T_{timestep})]^2 \quad (6)$$

2.2 Training Configuration Phase

In the training configuration phase, the timestep and neuron counts are determined from the estimated harvested energy $E_{harvested}$ and derived formulas. For a minor energy deficit, timestep adaptation is applied to preserve accuracy with limited savings, while a major deficit triggers neuron pruning for greater energy reduction at some performance cost. If both techniques fail to meet the energy target, training data is reduced. The configured SNN is then trained under carbon-neutral conditions.

The number of timesteps is calculated by substituting $E_{harvested}$ into Eq. 2. An excessively low number of

$T_{timesteps}$ will lead to significant model accuracy loss. Thus, the lower bound T_{min} is defined. If $T_{timesteps} < T_{min}$, $T_{timesteps}$ is fixed to T_{min} , and neuron pruning is performed to further reduce the energy consumption.

The number of neurons is calculated by substituting $E_{harvested}$ into Eq. 3. If the calculated $N_{neurons}$ is lower than $N_{neurons,min}$, the number of training data N_{train} is adjusted to $N_{train} \times \frac{N_{neurons}}{N_{neurons,min}}$. Then, $N_{neurons}$ is fixed to $N_{neurons,min}$. Finally, the calculated $(T_{timesteps}, N_{neurons}, N_{train})$ is the output of the framework.

2.3 Inference Configuration Phase

Similar to training configuration, the numbers of timesteps and neurons are determined by substituting $E_{inference}$ into Eq. 6. However, as the model is pre-trained, neurons are pruned based on similarity.

Similarity-based neuron pruning aims to compress the network by retaining only neurons with unique contributions. All possible neuron pairs are first generated, and their similarities are evaluated using an error-based metric such as Mean Squared Error (MSE). The pairs are then sorted in descending order of similarity, and redundant neurons from the most similar pairs are progressively removed to achieve efficient neuron compression while preserving model accuracy.

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2$$

Where:

- N represents the total number of weights in each neuron.
- x_i and y_i are the corresponding elements in the weight matrices.

After the configuration of $T_{timestep}$ and $N_{neurons}$, the SNN model performs inference.

3 Evaluation

3.1 Evaluation Methodology

The framework is implemented in Python using the BindsNet library [17] for model training and evaluation. For the energy estimation, open-sourced quartz-solar-forecast [14] is used. The amount of energy harvested is estimated based on cumulative Wh for 24h, 1-Wp panel, latitude: 51.75, longitude: -1.25 (Oxford, UK). For the amount of energy consumed, we perform energy analyses on Raspberry Pi 5 with a Cortex A7 CPU and 8GB LPDDR4X-4267 SDRAM, as well as using the energy

measurement tools YEREADW Type-C Tester. Additionally, we used Battery-dataset-preprocessing-code-library [15] to simulate the battery degradation.

3.2 Training Energy Adaptation

For this evaluation, we will examine nine scenarios that vary in terms of weather and battery conditions. The system optimization framework introduced in Section 2 will find $T_{timestep}$ and $N_{neurons}$ that satisfy an energy constraint for net-zero training. We also perform both single-node learning and ensemble learning [18] for a set of nodes with different conditions.

3.2.1 Single-node on-chip STDP learning

Table 1 shows the various energy conditions, adjusted configuration, and accuracy with respective configurations. A single 700-neuron model, which performs at 90.57% by default, is used. For a moderate energy deficit (0.00–15.15%), only the number of timesteps $T_{timestep}$ is adjusted, resulting in the marginal accuracy losses. For a significant energy deficit (88.10%), both the $T_{timestep}$ and $N_{neurons}$ are adjusted, resulting in a substantial accuracy loss. For all the energy conditions, the adapted consumptions are lower than the harvested energies, indicating the carbon-neutrality.

3.2.2 Ensemble on-chip STDP learning with a set of nodes

Training energy adaptation is further evaluated with the Ensemble STDP method in [18]. Table 2 summarizes the energy conditions for a single node with 70 neurons. A set of 10 nodes will be later merged to form a final model. For simplicity, the fresh battery condition is assumed for all the nodes. Similar to the single model case, only $T_{timestep}$ is adjusted for the 7.60% energy deficit case. However, for 88.21% energy deficit case, all $T_{timestep}$, $N_{neurons}$, and N_{train} are adjusted. This indicates that the calculated number of neurons did not satisfy the lower bound, and it is necessary to reduce N_{train} while $N_{neurons}$ is fixed at the lower bound.

Based on these configurations, ensemble learning with 10 nodes under three different weather conditions, and mixed weather conditions is performed [18]. Table 3 shows the corresponding accuracies of the single nodes and the merged final model. Only the slight accuracy drops can be observed for cloudy nodes and their merged model, where only $T_{timestep}$ is adjusted. For rainy nodes and their merged model, the accuracies drop compared to sunny and cloudy nodes. Similar performance is obtained for the mixed weather that contains rainy nodes.

3.3 Inference Energy Adaptation

Here, the inference energy adaptation of the framework is evaluated. Table 4 summarizes several energy deficit cases. Since the harvested energy is directly mapped to $T_{timestep}$ and $N_{neurons}$, both variables are always adjusted. Less than 1% of accuracy drop can be observed up to 77.49% energy deficit, and the highest accuracy drop is less than 10% while reducing 88.32% of inference energy. For the 27.18% energy deficit case, the adapted consumption exceeds the harvested energy.

4 Conclusion

In conclusion, the proposed *GreenMorph* framework achieves energy-efficient neuromorphic computing by integrating energy optimization and similarity-based neuron pruning. Experiments show that timestep adaptation reduces training energy by 15.15% without accuracy loss, while combining it with neuron pruning achieves an 88.10% reduction with only 3.01% accuracy loss. During inference, pruning cuts computation energy by up to 88.32% with minimal accuracy degradation. Future work will implement the framework on neuromorphic hardware for further energy reduction.

Table 1: Training Energy Adaptation to Various Energy Conditions in a single node (Raspberry Pi 5; network configuration: 784-700 with lateral inhibitory for winner-take-all mechanism; On-chip STDP learning).

Weather	Battery	Harvest (Wh)	Consumption (Wh)	Energy Deficit (%)	Adjusted Variables	Adapted Consumption (Wh)	Accuracy
sunny	fresh	174.24	172.50	0.00	None	174.24	0.9057
sunny	mid	160.30	172.50	7.07	$T_{timestep}$	156.10	0.9051
sunny	wear-out	146.36	172.50	15.15	$T_{timestep}$	140.00	0.9093
cloudy	fresh	159.12	172.50	7.76	$T_{timestep}$	158.90	0.9050
cloudy	mid	159.12	172.50	7.76	$T_{timestep}$	158.90	0.9050
cloudy	wear-out	146.36	172.50	15.15	$T_{timestep}$	140.00	0.9093
rainny	fresh	20.52	172.50	88.10	$T_{timestep}, N_{neurons}$	19.95	0.8756
rainny	mid	20.52	172.50	88.10	$T_{timestep}, N_{neurons}$	19.95	0.8756
rainny	wear-out	20.52	172.50	88.10	$T_{timestep}, N_{neurons}$	19.95	0.8756

Table 2: Training Energy Adaptation of a single node for Ensemble STDP (Raspberry Pi 5; network configuration: 784-70 with lateral inhibitory for winner-take-all mechanism; On-chip STDP learning).

Weather	Battery	Harvest (Wh)	Consumption (Wh)	Energy Deficit (%)	Adjusted Variables	Adapted Consumption (Wh)
sunny	fresh	2.66	2.63	0.00	None	2.63
cloudy	fresh	2.43	2.63	7.60	$T_{timestep}$	2.31
rainny	fresh	0.31	2.63	88.21	$T_{timestep}, N_{neurons}, N_{train}$	0.30

Table 3: Ensemble STDP Accuracy Comparison Under Different Weather Conditions (10 nodes, each node Raspberry Pi 5; network configuration: 784-(70x10) with lateral inhibitory for winner-take-all mechanism; On-chip STDP learning).

Node	Sunny Nodes	Cloudy Nodes	Rainy Nodes	Sunny Node: 30% Rainy Node: 70%	Cloudy Node: 30% Rainy Node: 70%	Rainy Node: 30% Sunny Node: 70%
0	0.7695	0.8120	0.6116	0.7695 (sunny)	0.8120 (cloudy)	0.6116 (rainy)
1	0.8056	0.7651	0.5959	0.8056 (sunny)	0.7651 (cloudy)	0.5959 (rainy)
2	0.8076	0.8155	0.6248	0.8076 (sunny)	0.8155 (cloudy)	0.6248 (rainy)
3	0.7777	0.7634	0.6276	0.6276 (rainy)	0.6276 (rainy)	0.7777 (sunny)
4	0.8143	0.7830	0.6103	0.6103 (rainy)	0.6103 (rainy)	0.8143 (sunny)
5	0.8065	0.8011	0.6575	0.6575 (rainy)	0.6575 (rainy)	0.8065 (sunny)
6	0.7742	0.7859	0.6444	0.6444 (rainy)	0.6444 (rainy)	0.7742 (sunny)
7	0.7993	0.7939	0.5659	0.5659 (rainy)	0.5659 (rainy)	0.7993 (sunny)
8	0.8208	0.7945	0.6802	0.6802 (rainy)	0.6802 (rainy)	0.8208 (sunny)
9	0.7572	0.7754	0.6304	0.6304 (rainy)	0.6304 (rainy)	0.7572 (sunny)
Node 0-9 merged	0.8536	0.8426	0.7279	0.7183 0.8317 (rainy excluded)	0.7181 0.8361 (rainy excluded)	0.6787 0.8444 (rainy excluded)

Table 4: Dynamic Pruning Energy Reduction During Inference (Single node, Raspberry Pi 5; network configuration: 784-700 with lateral inhibitory for winner-take-all mechanism).

Harvest (Wh)	Consumption (Wh)	Energy Deficit (%) (Wh, %)	Adjusted Variables	Accuracy	Adapted Consumption
9.75	9.42	0	None	0.8709	9.42
6.86	9.42	27.18%	$T_{timestep}, N_{neurons}$	0.8699	7.10
4.26	9.42	54.77%	$T_{timestep}, N_{neurons}$	0.8682	3.60
2.12	9.42	77.49%	$T_{timestep}, N_{neurons}$	0.8628	1.90
1.84	9.42	80.46%	$T_{timestep}, N_{neurons}$	0.8573	1.70
1.40	9.42	85.13%	$T_{timestep}, N_{neurons}$	0.8551	1.40
1.10	9.42	88.32%	$T_{timestep}, N_{neurons}$	0.7828	1.10

References

4336, 2020.

- [1] K. Muhammad *et al.*, “Deep learning for safe autonomous driving: Current challenges and future directions,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 7, pp. 4316–
- [2] U. Gupta *et al.*, “Chasing carbon: The elusive environmental footprint of computing,” in *2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2021, pp. 854–

- 867.
- [3] C.-J. Wu *et al.*, "Sustainable AI: Environmental implications, challenges and opportunities," *Proceedings of Machine Learning and Systems*, vol. 4, pp. 795–813, 2022.
 - [4] A. B. Abdallah and K. N. Dang, "Neuromorphic computing principles and organization," 2022.
 - [5] M. Nagel *et al.*, "A white paper on neural network quantization," *arXiv preprint arXiv:2106.08295*, 2021.
 - [6] D. Blalock *et al.*, "What is the state of neural network pruning?" *Proceedings of machine learning and systems*, vol. 2, pp. 129–146, 2020.
 - [7] R. Kobayashi *et al.*, "An efficient hardware implementation of spiking neural network using approximate Izhikevich neuron," in *2024 9th International Conference on Integrated Circuits, Design, and Verification (ICDV)*. IEEE, 2024, pp. 13–18.
 - [8] S. Han *et al.*, "Deep compression: Compressing deep neural networks with pruning, trained quantization, and huffman coding," *International Conference on Learning Representations (ICLR)*, 2015, arXiv:1510.00149.
 - [9] N.-D. Nguyen *et al.*, "Power-Aware Neuromorphic Architecture With Partial Voltage Scaling 3-D Stacking Synaptic Memory," *IEEE Trans. VLSI Syst.*, vol. 31, no. 12, pp. 2016–2029, 2023.
 - [10] Q. Wu *et al.*, "Clock-Gating and Its Application to Low Power Design of Sequential Circuits," *IEEE Trans. Circuits Syst. I*, vol. 47, no. 3, pp. 415–420, 2000.
 - [11] N.-D. Nguyen *et al.*, "An in-situ dynamic quantization with 3D stacking synaptic memory for power-aware neuromorphic architecture," *IEEE Access*, vol. 11, pp. 82 377–82 389, 2023.
 - [12] T. Kolpe *et al.*, "Enabling Improved Power Management in Multicore Processors through Clustered DVFS," in *2011 Design, Automation & Test in Europe*. IEEE, 2011, pp. 1–6.
 - [13] H. Ritchie *et al.*, "Data page: Carbon intensity of electricity generation," <https://archive.ourworldindata.org/20250624-125417/grapher/carbon-intensity-electricity.html>, 2023, part of the publication "Energy". Data adapted from Ember and Energy Institute. Archived on June 24, 2025. Licensed under CC BY.
 - [14] P. Dudfield *et al.*, "open-source-quartz-solar-forecast," Mar. 2025. [Online]. Available: <https://github.com/openclimatefix/open-source-quartz-solar-forecast>
 - [15] F. Wang *et al.*, "Physics-Informed Neural Network for Lithium-Ion Battery Degradation Stable Modeling and Prognosis," *Nature Communications*, vol. 15, no. 1, p. 4332, 2024.
 - [16] P. U. Diehl and M. Cook, "Unsupervised learning of digit recognition using spike-timing-dependent plasticity," *Frontiers in computational neuroscience*, vol. 9, p. 99, 2015.
 - [17] H. Hazan *et al.*, "Bindsnet: A machine learning-oriented spiking neural networks library in python," *Frontiers in neuroinformatics*, vol. 12, p. 89, 2018.
 - [18] H. Yuga and K. N. Dang, "EnsembleSTDP: Distributed in-situ spike timing dependent plasticity learning in spiking neural networks," in *2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc)*. IEEE, 2024, pp. 434–440.